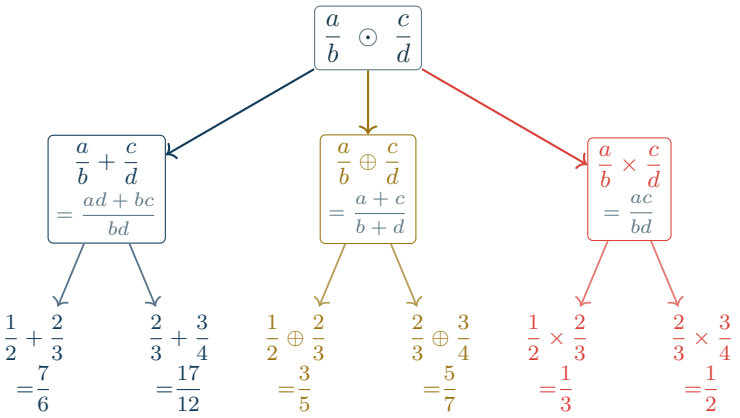
ARITHMETIC ON ONE PAGE

A ternary tree of operations on fractions

Sum, mediant and product are three ways to combine two fractions, and polynomials explain why there are exactly three.

1 / THE RULE AND THE PICTURE

Where a binary tree has two branches, a tree for fractions has three



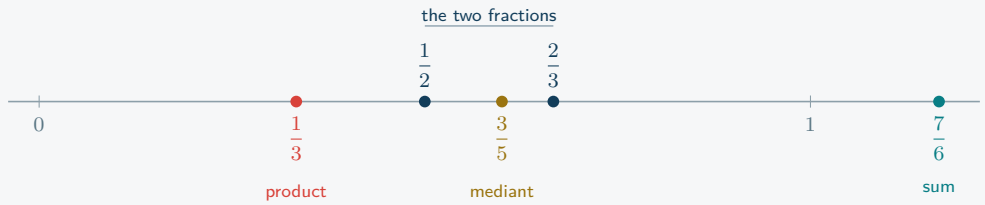
The symbol $\odot$ stands for any way of combining two fractions. Three are in common use:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$$
$$\frac{a}{b} \oplus \frac{c}{d} = \frac{a+c}{b+d}$$
$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

The leaves use neighbouring fractions $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}$, as the book uses $1, 2, 3$.

2 / WHERE THE THREE RESULTS LAND

The mediant stays between the two fractions, and the sum and the product leave



	sum $+$	mediant $\oplus$	product $\times$
commutative	yes	yes	yes
associative	yes	yes, for fixed numerators and denominators	yes
identity	0	none	1
depends only on the values, $\frac{1}{2} = \frac{2}{4}$	yes	no: $\frac{1}{2} \oplus \frac{1}{3} = \frac{2}{5}$ but $\frac{2}{4} \oplus \frac{1}{3} = \frac{3}{7}$	yes
where the result lies	above both	between the two	below both, for fractions under 1
from $\frac{1}{2}$ and $\frac{2}{3}$	$\frac{7}{6}$	$\frac{3}{5}$	$\frac{1}{3}$
from the polynomials of panel 3	multiply: middle coefficient	add	multiply: last coefficient

3 / FRACTIONS AS ROOTS OF LINEAR POLYNOMIALS

Two operations on polynomials make three operations on fractions

Write $\frac{a}{b}$ as the linear polynomial $bx - a$, whose root is $\frac{a}{b}$. Adding two of them gives $(b+d)x - (a+c)$, whose root is the mediant. Multiplying them gives a quadratic whose two roots are $\frac{a}{b}$ and $\frac{c}{d}$, so Vieta's formulas read off the sum and the product of the fractions from its coefficients:

$$(bx - a)(dx - c) = bd\,x^2 - (ad + bc)\,x + ac$$

The sum is the middle coefficient over the first, $\frac{ad+bc}{bd}$, and the product is the last over the first, $\frac{ac}{bd}$. For $\frac{1}{2}$ and $\frac{2}{3}$ we get $(2x-1)(3x-2) = 6x^2 - 7x + 2$, with roots adding to $\frac{7}{6}$ and multiplying to $\frac{2}{6} = \frac{1}{3}$, while $(2x-1) + (3x-2) = 5x - 3$ has the root $\frac{3}{5}$. The middle coefficient $ad + bc$ is the permanent of the array whose determinant $ad - bc$ is ± 1 for Farey neighbours such as $\frac{1}{2}, \frac{2}{3}$.

4 / CLIMBING THE LEVELS OF ABSTRACTION

The analogies change as the symbols become more general

At the first level $\frac{1}{2} + \frac{2}{3}$ is analogous to $\frac{2}{3} + \frac{3}{4}$, two sums of neighbours, but not to $\frac{1}{2} \times \frac{2}{3}$. Replacing the numbers by $\frac{a}{b}$ and $\frac{c}{d}$ lets every leaf of a branch be treated alike, and replacing the operation by $\odot$ lets the three branches be compared. Now the sum is analogous to the product, since both are commutative and associative with an identity (0 and 1), and the product distributes over the sum, while the mediant stands apart because it depends on how a fraction is written. One further pivot, to polynomials, shows the mediant as ordinary addition and the sum and product as ordinary multiplication, so the three branches come from the two operations of a ring.

ASIDE / TRY THESE

For $\frac{1}{3}$ and $\frac{3}{4}$ the sum is $\frac{13}{12}$, the mediant is $\frac{4}{7}$ and the product is $\frac{1}{4}$; check them with $(3x-1)(4x-3) = 12x^2 - 13x + 3$ and $(3x-1) + (4x-3) = 7x - 4$.

LINKS TO OTHER SHEETS

The mediants and double angles sheet treats $\oplus$ and Farey neighbours; the partitions sheet uses the same habit of climbing a level.

Convention. $\oplus$ is the mediant; $\odot$ any binary operation; a/b and c/d are written in lowest terms with positive entries; all values were checked with exact fractions.

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